Generative Ai And Ai Platform Engineering

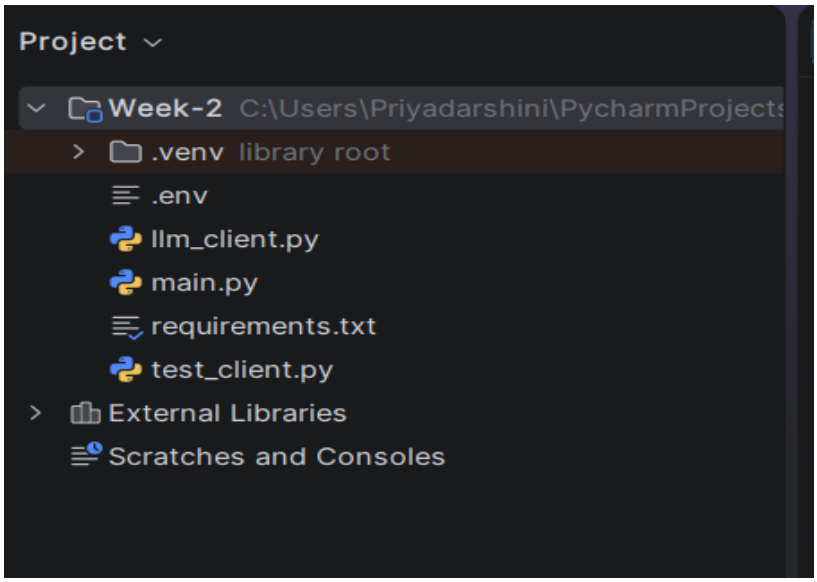
Name:- K. Priyadarshini

Roll no:- 2520030067

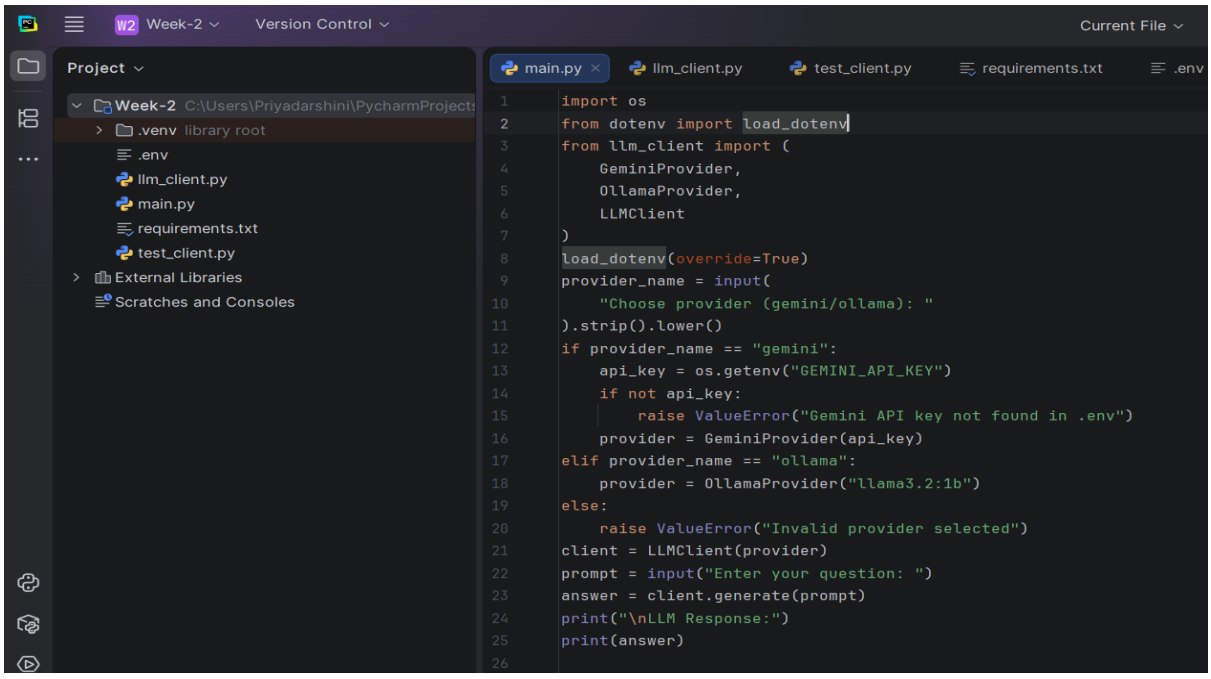
Section no:- 3

WEEK-2

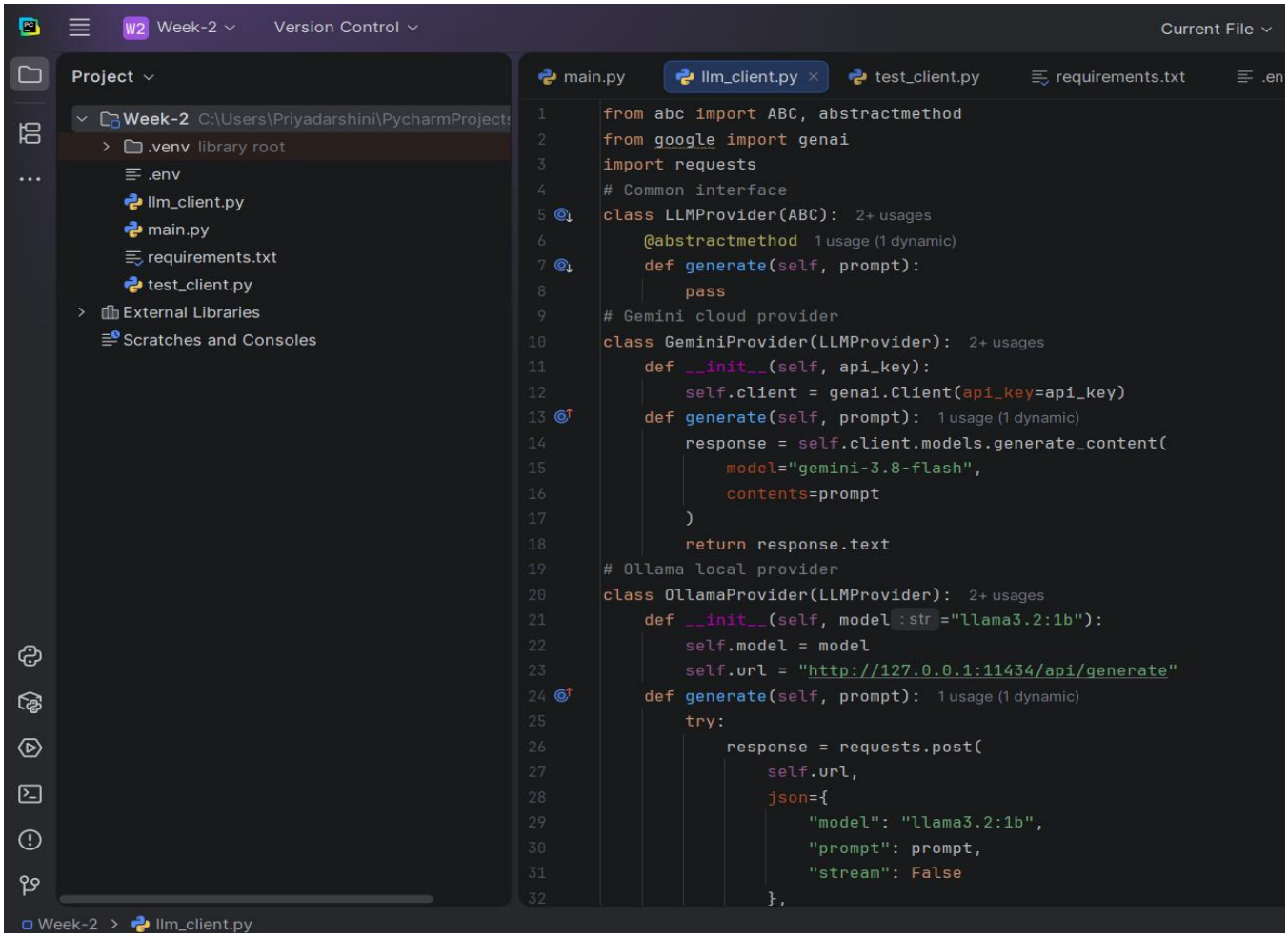
Step 1:-



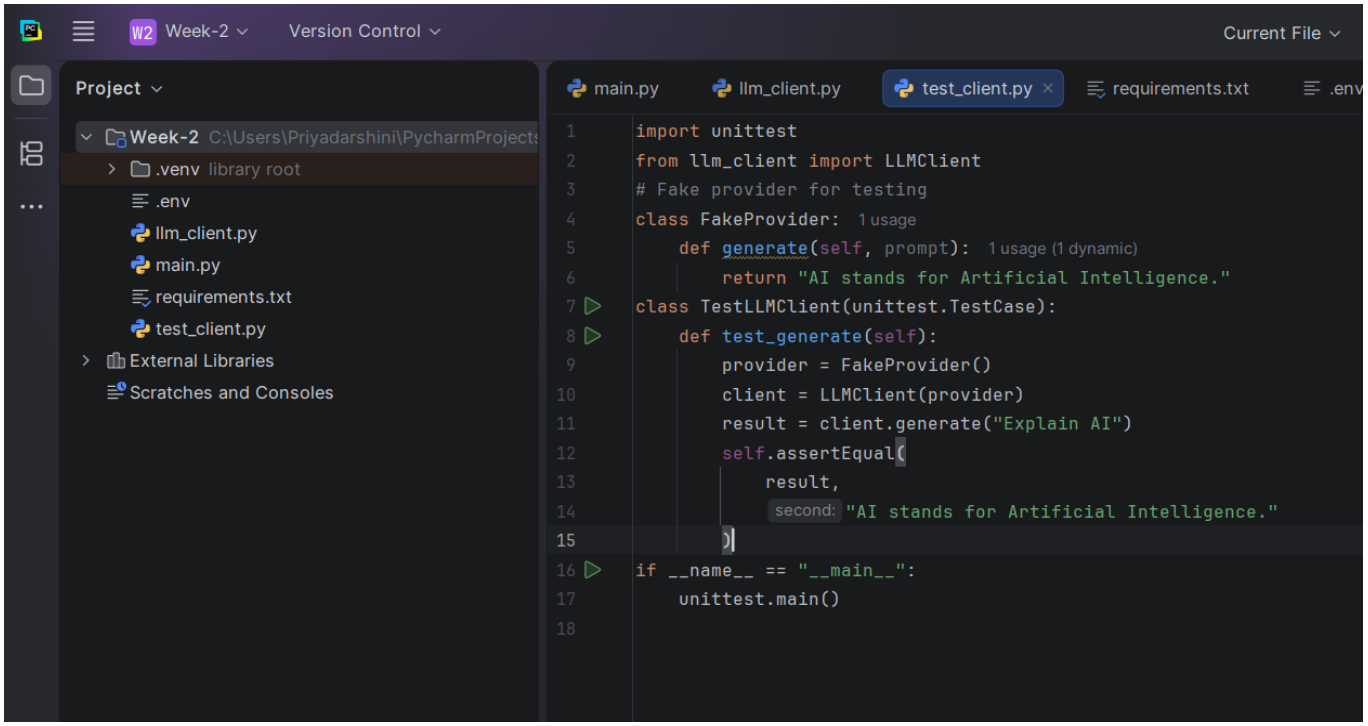
Step 2:-



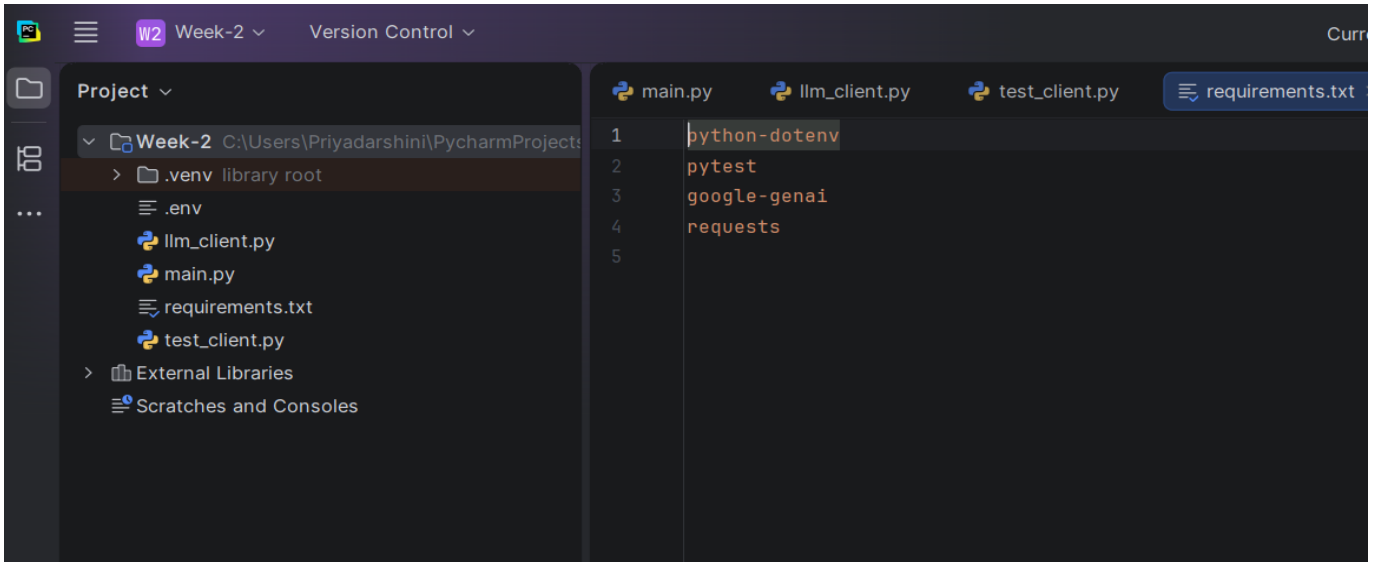
Step 3:-


Week-2
Version Control
Current File
Project
Week-2 C:\Users\Priyadarshini\PycharmProjects
.venv library root
.env
llm_client.py
main.py
requirements.txt
test_client.py
External Libraries
Scratches and Consoles
main.py
llm_client.py
test_client.py
requirements.txt
from abc import ABC, abstractmethod
from google import genai
import requests
Common interface
class LLMProvider(ABC): 2+ usages
@abstractmethod 1 usage (1 dynamic)
def generate(self, prompt):
pass
Gemini cloud provider
class GeminiProvider(LLMProvider): 2+ usages
def __init__(self, api_key):
self.client = genai.Client(api_key=api_key)
def generate(self, prompt): 1 usage (1 dynamic)
response = self.client.models.generate_content(
model="gemini-3.8-flash",
contents=prompt
)
return response.text
Ollama local provider
class OllamaProvider(LLMProvider): 2+ usages
def __init__(self, model: str ="llama3.2:1b"):
self.model = model
self.url = "http://127.0.0.1:11434/api/generate"
def generate(self, prompt): 1 usage (1 dynamic)
try:
response = requests.post(
self.url,
json={
"model": "llama3.2:1b",
"prompt": prompt,
"stream": False
},
Week-2
llm_client.py

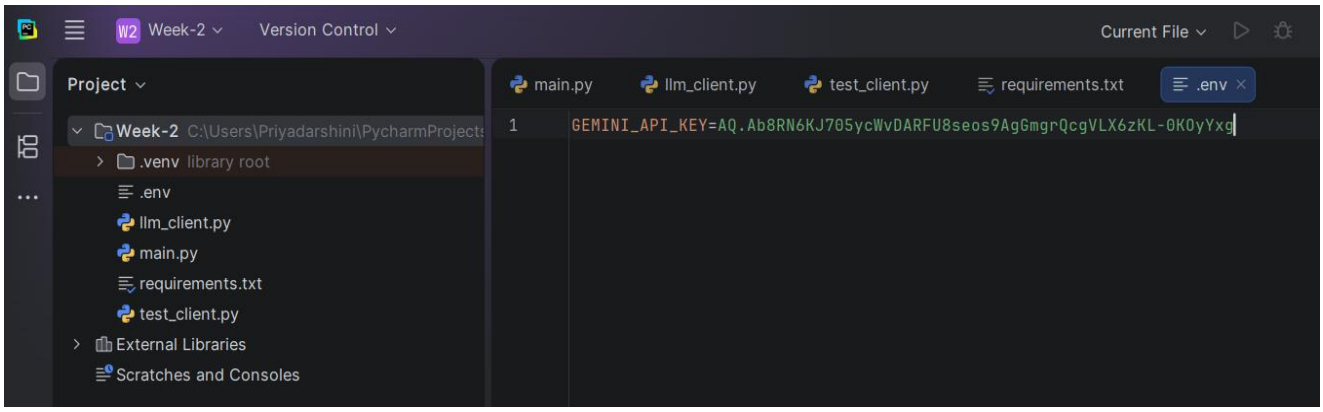
Step 4:-


Week-2
Version Control
Current File
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Week-2 C:\Users\Priyadarshini\PycharmProjects
.venv library root
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External Libraries
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test_client.py
requirements.txt
import unittest
from llm_client import LLMClient
Fake provider for testing
class FakeProvider: 1 usage
def generate(self, prompt): 1 usage (1 dynamic)
return "AI stands for Artificial Intelligence."
class TestLLMClient(unittest.TestCase):
def test_generate(self):
provider = FakeProvider()
client = LLMClient(provider)
result = client.generate("Explain AI")
self.assertEqual(
result,
second: "AI stands for Artificial Intelligence."
)
if __name__ == "__main__":
unittest.main()

Step 5:



Step 6:-



OUTPUT:-

